Research

Plastic tooth implants with root channels and osseous bridges

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The clinical success of the plastic tooth implant was initially reported by Hodosh,[1, 2] and subsequent reports have established the validity of this technique, with the use of a number of different procedures in a variety of experimental animals, primarily dogs, monkeys, and baboons.[2-5]

The plastic teeth, exact replicas of extracted natural teeth, were constructed of polymethylmethacrylate and were fashioned immediately upon extraction of a natural tooth so that the plastic implant could be placed in the natural socket within 30 minutes after removal of the natural tooth. Adaptation to the socket was easily attained since the plastic tooth was an exact replica of the extracted tooth. It was found that the initial success of the implant depended to a certain extent upon the absence of occlusal traumatizing influences. Because of the powerful bite in baboons, the plastic implant teeth were secured by fixation, either by splinting to adjacent teeth with stainless steel wire and acrylic or by a technique of intrabony pinning, whereby a Vitallium wire pin was inserted through the buccal plate of bone, through the root of the implant, and into the lingual or palatal plate of alveolar bone.

The pinning technique was shown to be highly successful in securing the implant tooth over a long period of time.

Histologic investigations revealed that the blood clot eventually developed into fibrous connective tissue and the plastic implant tooth was surrounded by dense collagen fibers analogous to the periodontal membranes surrounding the roots of natural teeth. The periodontal membrane serves to attach the natural tooth to the alveolar bone of the socket and allow for functional movement of the tooth. The periodontal fibers run from the cementum of the tooth to the alveolar

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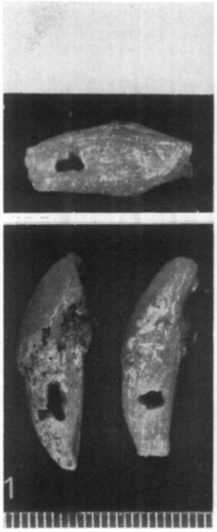

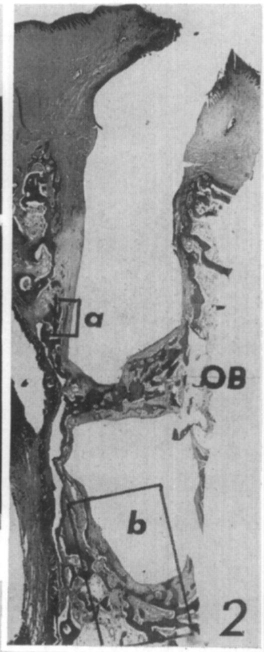

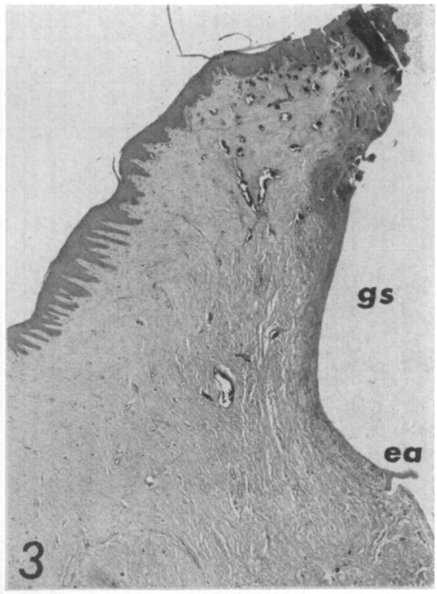

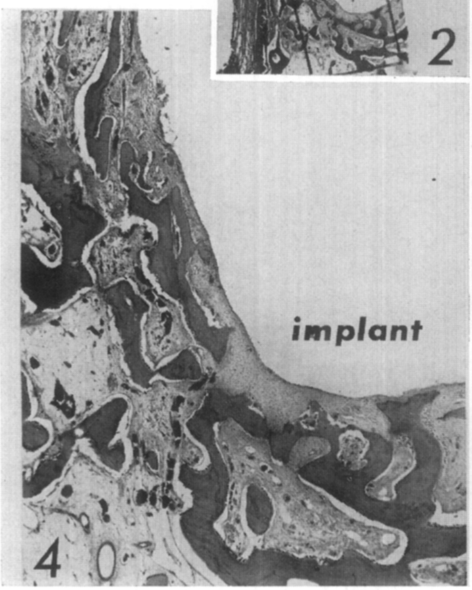

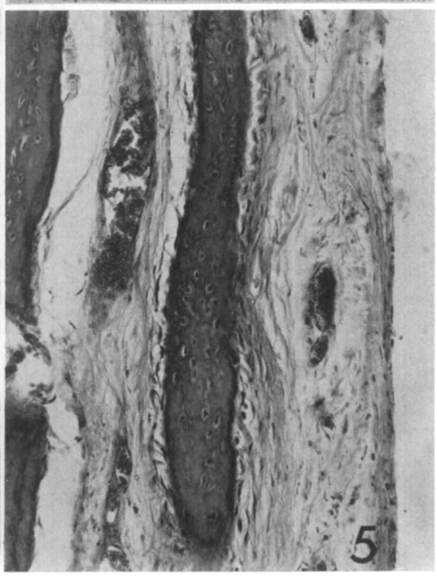

Fig. 1. Three plastic implant teeth inserted into baboon. Note large holes that were drilled through middle of roots. Millimeter scale gives size of openings.

Fig. 2. Survey section of plastic implant tooth. Horizontal osseous bridge *(OB)* running in mesiodistal direction is clearly discerned. (Hematoxylin and eosin stain. Magnification, ×30.)

Fig. 3. Marginal gingiva shown in Fig. 2, demonstrating minimal inflammatory infiltration in relation to gingival sulcus *(gs)*. *Base* of epithelial attachment is seen at position *ea*. Apical proliferation of epithelial attachment is minimal. (Hematoxylin and eosin stain.) (Magnification, ×100.)

Fig. 4. Medium-power magnification of Fig. 2, area *b*, showing periodontium at apex of plastic tooth implant. Regular alveolar bone is noted. (Hematoxylin and eosin stain. Magnification, ×100.)

Fig. 5. High-power view of Fig. 2 area showing periodontal membrane facing plastic tooth implant. Periodontal membrane is composed of collagen bundles, and adjacent alveolar bone presents osteoblastic activity. (Hematoxylin and eosin stain. Magnification, ×200.)

bone and generally are horizontal or oblique in orientation except at the apex of the root, where the orientation tends toward the vertical. Osteoblastic activity is usually noted along the alveolar bone. In the plastic tooth implant, the surrounding alveolar bone was found to present osteoblastic activity along the periodontal membrane surface. The fibers in the periodontal membrane surrounding plastic implant teeth tended to run in a more vertical orientation. Attachment of fibers to the plastic surface was surmised but could not be definitively established because of the difficulty of preparing histologic sections of the plastic tooth in situ. This problem has been overcome recently, and fibers of the periodontal membrane were seen to insert directly into the plastic material.[6]

Although the plastic implant teeth remained firm after removal of occlusal splints, and could even be used as bridge abutments, it was thought that, if bony bridges could be made to grow through large spaces in the root, the plastic tooth would be more firmly secured.

In order to test the validity of this hypothesis and the possibility of developing bony bridges through the plastic roots, a number of implants were devised, with large holes prepared through the root (Fig. 1) ranging from 1 to 2.5 mm. in diameter.

MATERIALS STUDIED

Six plastic implant teeth with wide channels cut through the roots were studied. The channels were prepared mesiodistally in a horizontal direction. As in previous studies, six natural teeth in two baboons *(Papio doguera)* were selected for plastic tooth replacement. The teeth selected were four maxillary central incisors, a maxillary right lateral incisor, and a right mandibular central incisor. The animals were tranquilized and anesthetized with intravenous pentobarbital. The teeth were extracted and replicated in polymethylmethacrylate. The plastic replicas were fashioned and inserted into the animals' natural alveolar sockets within 30 minutes after extraction of the natural teeth. Prior to insertion of the plastic tooth implants, the horizontal channels, 1 to 2.5 mm., were drilled through the roots. The plastic teeth were splinted to adjacent natural teeth with 26 gauge braided stainless steel wire covered with self-curing acrylic resin. The splints were intracoronal and were placed in grooves cut on the occlusal surfaces of the splinting abutment tooth and the implant. The techniques of plastic tooth fabrication, placement, and splinting have been described in a previous publication.[5]

RESULTS

Gross observations

The plastic tooth implants, each with horizontal channels cut through the root, were extremely firm clinically after periods of from 6 to 12 months after initial placement. At the time of tissue preparation for microscopic observation, the implants were difficult to move within the socket. After decalcification of tissue blocks, the implants were firmly held in position and could be removed only by vertical incisions and sectioning of one end of the horizontal tissue root bridge.

Fig. 6. Medium-power magnification of osseous bridge (OB) through prepared horizontal root channel. Formation of mature bone is noted, with connective tissue superimposed [illegible] to bone. Hematoxylin and eosin stain. Magnification, ×100.

Fig. 7. High-power magnification of Fig. 6, area x. Mature bone with osteoblastic [illegible] at margins is noted. Hematoxylin and eosin stain. Magnification, ×300.

Fig. 8. High-power magnification of Fig. 6, area y, showing fibrous connective tissue and trabeculae of bone with osteoblastic activity. Hematoxylin and eosin stain. Magnification, ×300.

Fig. 9. High-power magnification of Fig. 6, area z, showing new bone formation at inferior border of osseous bridge. Intense osteoblastic activity is noted, and [illegible]. Hematoxylin and eosin stain. Magnification, ×300.

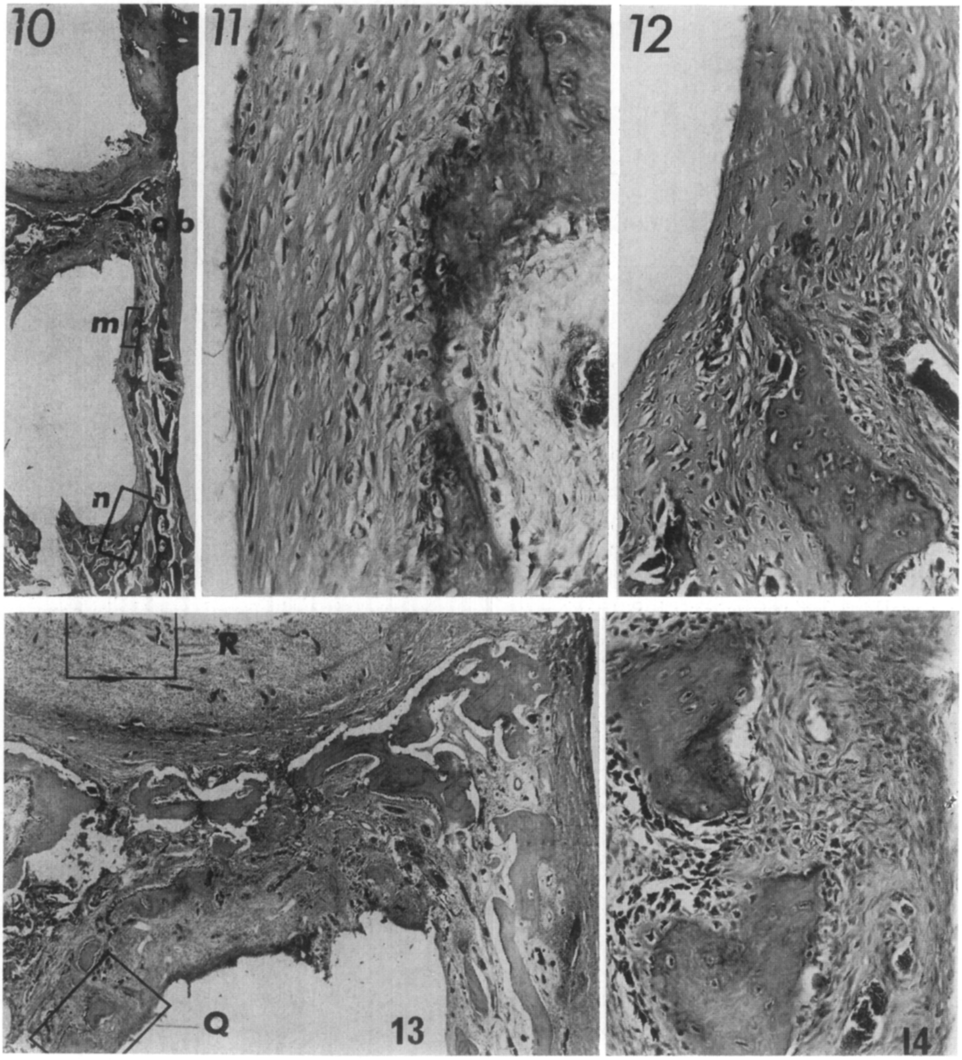

Fig. 10. Survey section of plastic implant with horizontal channel through root and osseous bridge *(ob)*. Tissue tear occurs during removal of implant prior to histologic preparation. (Hematoxylin and eosin stain. Magnification, ×30.)

Fig. 11. High-power view of lateral periodontal membrane (Fig. 10, area *m*). Periodontal connective tissue fibers follow vertical and oblique orientation. Osteoblastic activity is evident. (Hematoxylin and eosin stain. Magnification, ×350.)

Fig. 12. High-power view of apical periodontium (Fig. 10, area *n*). Active osteogenesis is seen, with osteoid tissue and beadlike patterns of osteoblasts. (Hematoxylin and eosin stain. Magnification, ×300.)

13. Low-power view of horizontal osseous bridge shown in Fig. 10. Osseous trabeculae are noted, but inflammatory infiltration is evident at *R*. (Hematoxylin and eosin stain. Magnification, ×100.)

Fig. 14. High-power view of Fig. 13, area *Q*. Trabeculae of newly formed bone are noted. (Hematoxylin and eosin stain. Magnification, × 350.)

Microscopic observations

The microscopic observations revealed well-developed osseous bridges that had developed through the horizontal channels cut through the roots of the plastic implants (Fig. 2). In those instances in which the gingival and periodontal tissues were well adapted to the plastic implant and in which gingival inflammation was minimal (Figs. 3, 4, and 5), the osseous root bridge was well developed and free of inflammatory infiltration (Figs. 6 and 7). The bone was well adapted to the plastic surface and was separated from the plastic by a band of connective tissue similar to a periodontal membrane structure (Figs. 8 and 9). Active osteogenesis was observed at this connective tissue membrane. Torn collagen bundles at the connective tissue surface were noted (Figs. 8 and 9). These fibers probably passed into the plastic itself and were torn off when the plastic material was removed before tissue preparation. In those cases where the root channel was in the coronal portion of the root and close to the gingival sulcus, osseous bridges developed but inflammatory infiltration could be noted, and this inflammation was continuous with gingival inflammation related to a deepened gingival sulcus or periodontal pocket (Fig. 10). In these instances, the alveolar bone and the periodontal membrane tissue apical to the osseous bridge were usually regular and free of inflammation (Figs. 11 and 12). The osseous bridges in these instances developed well-formed bony trabeculae along their radicular surfaces (Figs. 13 and 14) but demonstrated granulation tissue as well as inflammatory infiltration of loose connective tissue along their coronal surfaces.

DISCUSSION

The development of osseous bridges through horizontally placed root channels in plastic teeth is of considerable interest. The development of this type of structure would serve to anchor the implant more firmly within the socket. Our clinical observations in these animals were in accord with these histologic findings. The implants were extremely firm. The implants presented less of a periodontal inflammatory picture when the root channel was placed toward the apical portion of the root. In one instance where the channel was in the coronal portion of the root, an osseous bridge developed but there was an inflammatory infiltration extending from the gingival tissues, where gingivitis and a periodontal pocket were noted.

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